

Prompt Engineering Essentials

Topics

- e! Introduction to Prompt Engineering
- e! Structure and Elements of Prompts
- e! Zero-shot Prompting
- e! One-shot Prompting
- e! Few-shot Prompting
- e! Instruction Tuning Basics
- e! Prompt Testing and Evaluation
- e! Prompt Pitfalls and Debugging
- e! Prompts for Different NLP Tasks
- e! Understanding Model Behavior with Prompt Variations

Learning Objectives

By the end of this lesson, you will be able to:

- e! Understand the core principles, components, and use cases of prompt engineering
- e! Identify the structure and characteristics of effective prompts and templates.
- e! Apply zero-shot, one-shot, and few-shot prompting techniques to NLP tasks.
- e! Explain the role of instruction tuning and its relevance to prompt optimization.
- e! Evaluate prompt quality using testing methods and recognize common pitfalls.
- e! Design customized prompts using strategies and tools.

Introduction to Prompt Engineering

The Power of Prompt Engineering

Many individuals interact with AI regularly, often unaware that the manner in which queries are phrased directly impacts the relevance and depth of the responses.

The Power of Prompt Engineering

Many individuals interact with AI regularly, often unaware that the manner in which queries are phrased directly impacts the relevance and depth of the responses.

Alex: "Hey Sam, why did your AI give you a much better answer than mine?"



Sam: "Because I knew how to ask — it's called prompt engineering!"

What is a Prompt?

A prompt is a natural language text that requests the generative AI to perform a specific task.

ChatGPT 4o

Share

Prompt

Write a JavaScript function that takes a string as input and returns the reversed string

Here's a simple JavaScript function that takes a string as input and returns the reversed string:

```
javascript
function reverseString(str) {
  return str.split('').reverse().join('');
}

// Example usage
console.log(reverseString("hello")); // Output: "olleh"
```

Copy Edit



What is Prompt Engineering?

Prompt engineering is a process of designing and optimizing prompts to achieve optimal results from a generative artificial intelligence (AI) model.



Crafting and refining inputs



Translating human intent



Balancing creativity and logic

How does Prompt Engineering work?



Understand the task and define objectives before prompt creation.

01



Create a precise and specific prompt to guide the AI effectively.

02



Test the prompt to evaluate the quality of the AI's response.

03



Scale the optimized prompt for similar or broader use cases.

06



Review the AI-generated content for accuracy and relevance.

05



Refine the prompt iteratively based on output feedback.

04

Why Prompt Engineering?



Structure and Elements of Prompts

Core Elements of Prompt

Instruction : The specific task given to the AI to guide its response.

Context : Additional context that can steer the model for better responses.

Output Indicator : Specifies how the response should be structured.

Input Data : The Content, question, or information the AI processes.

Core Elements of Prompt (Contd.)

Example Prompt: Summarize the key findings and insights of the following research abstract in 3 professionally worded bullet points suitable for inclusion in a business report.

"The study analyzes the adoption of AI tools in mid-sized enterprises. Results indicate improved operational efficiency, cost reduction, and enhanced decision-making accuracy. However, challenges include skill gaps and integration issues."

Instruction: Summarize key findings and insights

Context: Intended for a professional business report

Input Data: The content of the research abstract

Output Indicator: 3 bullet points in professional tone

Additional Elements for Advanced Prompting

Element	Description	Example Prompt
Persona (Role)	Assigns a specific role or expertise to the AI.	You are a financial advisor. Suggest investment options for a risk-averse client.
Exemplar (Examples)	Provides sample inputs and outputs to guide the model.	Q: What is the capital of France? A: Paris. Q: What is the capital of Japan? A: Tokyo.
Tone and Constraints	Specifies style, tone, or limits for the response.	Write a formal email declining a meeting. Limit your response to 50 words.
Delimiters and Syntax	Uses markers to separate instructions, context, or data.	Summarize the text between the triple backticks: The quick brown fox jumps over the lazy dog.
Technique	Instructs the AI to use a reasoning or approach.	List the pros and cons of remote work. Then, provide a recommendation.

Example for Effective Prompting (Contd.)

Effective Prompt

“Write a professional email responding to a customer requesting a refund. Address their concern politely, ask for relevant details (like order number or purchase date), and provide steps to process the refund ”

Subject: Refund Request – Additional Information Required

Dear [Customer's Name],

Thank you for reaching out to us regarding your refund request. We're sorry to hear that you were not fully satisfied with your purchase, and we're here to assist you.

To help us process your refund smoothly, could you please provide the following details:

- Order number
- Date of purchase
- Reason for the refund request
- Any supporting documents (e.g., receipt, screenshots, or photos if applicable)

Once we receive this information, we will review your request and guide you through the next steps of the refund process. Please note that our standard processing time for refunds is [X business days] from the date we receive the required information.

We appreciate your patience and understanding, and we look forward to resolving this for you as quickly as possible.

Warm regards,

[Your Full Name]

[Your Position]

[Company Name]

[Contact Information]



Hard Prompts vs. Soft Prompts

Aspect	Hard Prompts	Soft Prompts
Definition	Human-written text instructions.	Learned embeddings, not human-readable.
Nature	Discrete language tokens.	Continuous vectors.
Interpretability	Easy to interpret.	Not interpretable.
Creation	Manually crafted.	Optimized during training.
Use Cases	General tasks, prototyping.	High-precision, domain-specific tasks.
Flexibility	General but manual.	Task-specific and scalable.
Example	<i>"Summarize this text."</i>	Fine-tuned embeddings for sentiment.
Advantages	Simple, clear, easy to use.	Precise, efficient, adaptable.
Disadvantages	Needs human effort, can vary in quality.	Less transparent, resource-intensive.

Prompt Engineering Use Cases



Prompting Techniques

What are Prompt Engineering Techniques?

Prompt engineering techniques refer to structured methods used to design, format, and refine inputs (prompts) given to generative AI models.



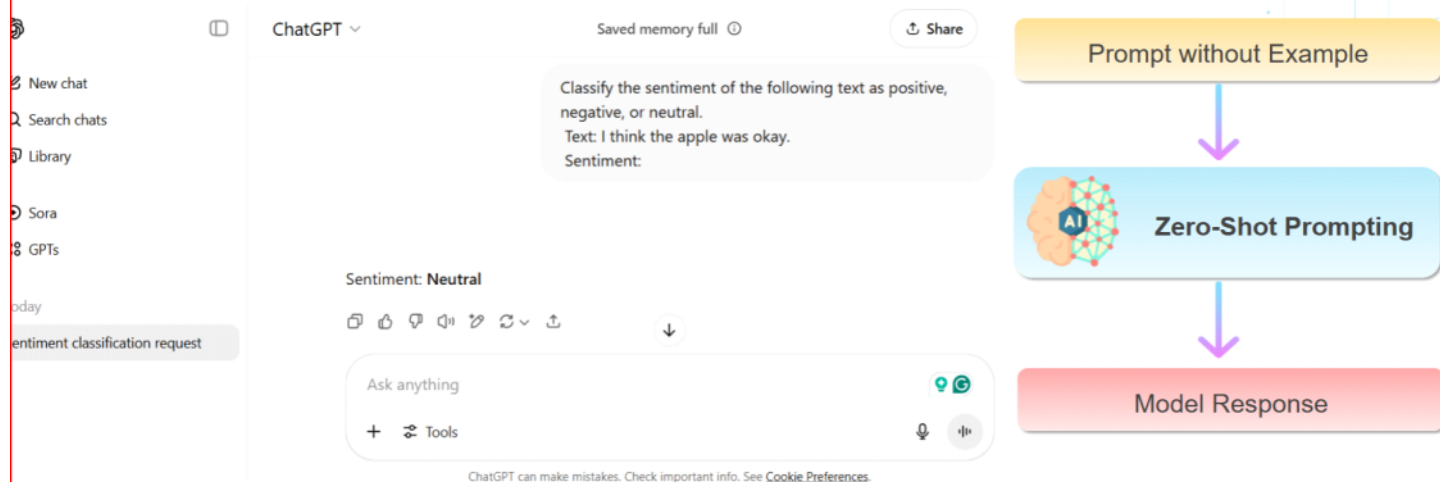
Zero-Shot Prompting

What is Zero-Shot Prompting?

Zero-shot prompting is the simplest form of prompting, which involves providing the model with a direct instruction to perform a task without any examples or demonstrations.



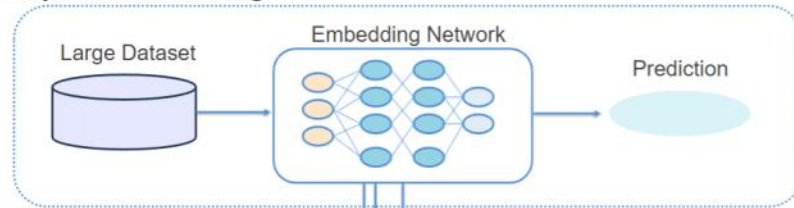
How Zero-Shot Prompting Works?



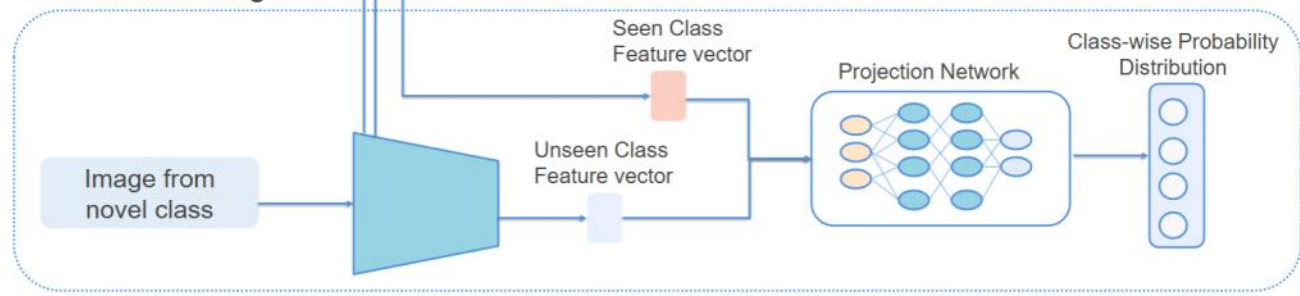
The lack of examples leaves the model guessing, and the results can be **unpredictable** or **incorrect**.

Zero-Shot Learning Process

Supervised Learning



Zero-Shot Learning



Pros and Cons of Zero-Shot Prompting

Versatile: Handles many tasks without retraining.



Efficient: No need for labelled data; faster results.



Adaptable: Tackles new tasks using prior knowledge.



Lower Accuracy: May miss nuances or task-specific detail.



Ambiguity: Can misread vague prompts.



Data-Dependent: Limited by gaps in training data.



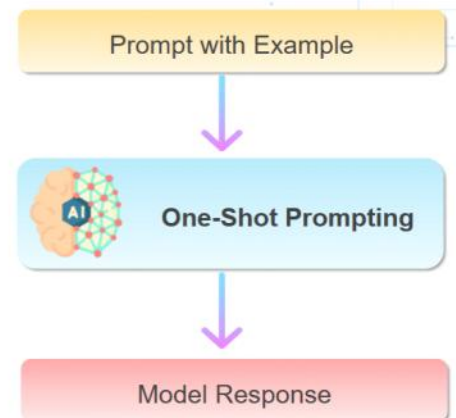
One-Shot Prompting

What is One-Shot Prompting?

One-shot prompting refers to the method where a model is provided with a single example or prompt to perform a task.



How One-Shot Prompting Works?



In this example, the single prompt guides the model in producing **accurate answers** by following the **provided format**.

Pros and Cons of One-Shot Prompting

Clear Guidance: Easy to understand with one example.



Improved Accuracy: Boosts relevant responses.



Resource Efficient: Saves data with one example.



Versatile: Works across tasks like translation.



Limited for Complex Tasks: Struggles with complex tasks.



Example Sensitive: Relies on quality of the single example.



Overfitting Risk: Might copy the example too closely.



Challenges with Novel Tasks: Struggles with tasks.

Few-Shot Prompting

What is Few-Shot Prompting?

Few-shot prompting is a technique where a large language model is given a small number of examples (typically 2–5) within the prompt to help it understand and perform a task.

A Few Shot Prompt

Example 1

Example 2

More Examples

Your Input

Classify these conversations

Example 1

Text: I forgot my password and can't log in.

Label: Account Issue

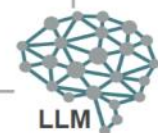
Example 2

Text: How do I change my billing address?

Label: Billing Query

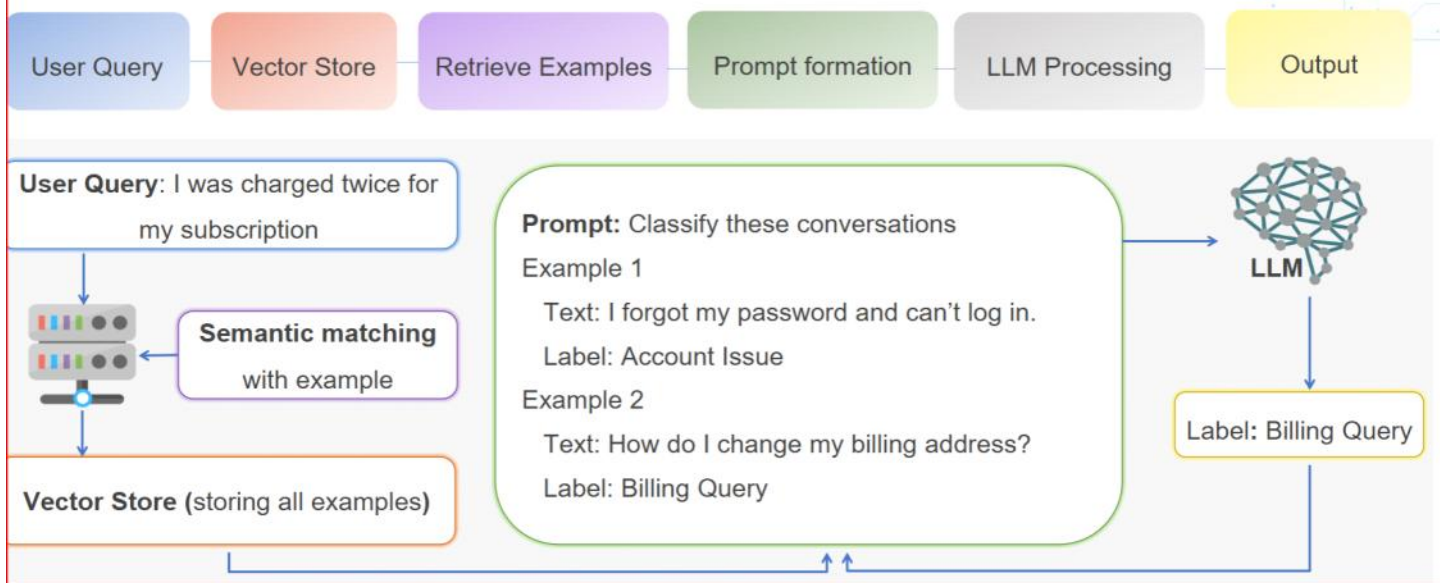
Input: I was charged twice for my subscription

Semantic matching with example



LLM

How does Few-Shot Prompting Works?



Pros and Cons of Few-Shot Prompting

Advantages



Data-Efficient: Works well with minimal labelled data.



Improved Accuracy: Boosts results in tasks like translation or classification.



Cost-Effective: Often avoids full retraining, saving time, and resources.

Limitations

Prompt Dependent: Needs well crafted examples to perform well.



High Compute: Large models require significant resources.



Limited Generalization: Struggles with complex or unfamiliar tasks.



How to Choose the Right Prompting

Criteria	Zero-Shot Prompting	One-Shot Prompting	Few-Shot Prompting
Accuracy	Lower	Moderate	Highest
Resource Requirements	Minimal	Moderate	Higher
Task Complexity	Simple	Moderate	Complex
Generalization	Model knowledge	One example	Multiple examples
Prompt Effort	Very low	Low	Moderate
Scalability	High	High	Lower
Adaptability	Low	Moderate	High

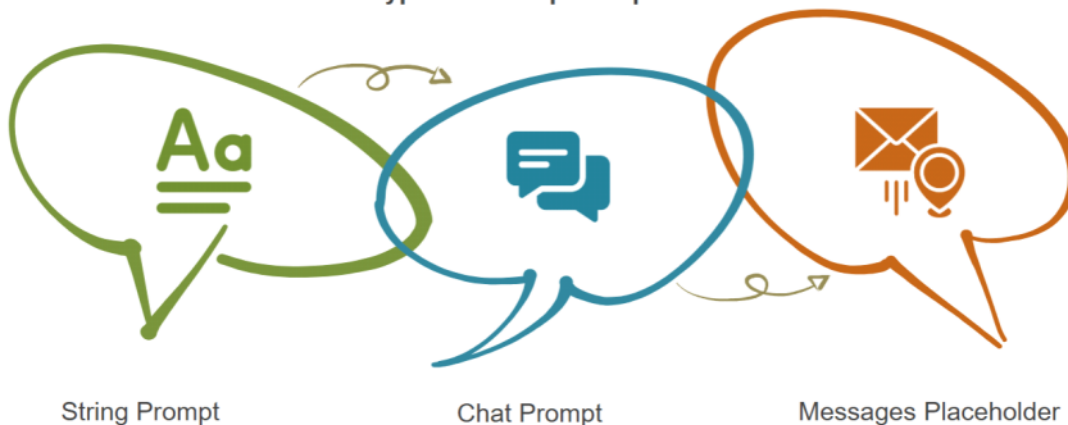
Demo 1: Crafting Effective Prompt Techniques

Prompt Templates

What are Prompt Templates?

A **prompt template** is a pre-defined structure or blueprint used to construct prompts for language models.

Types of Prompt Template



String Prompt Template



Definition

Single-line prompts with variables embedded.

Use Case

Best for direct tasks like Q&A or summarization.

Example

Tell me a joke about {topic} → Tell me a joke about cats.

Chat Prompt Template



Structure

Organizes messages by roles (system and assistant).

Customization

Variables can be added to each message.

Application

Ideal for simulating realistic conversations.

Messages Placeholder



Function

Adds dynamic message history into a template.

Purpose

Maintains continuity and context.

Usage

Used in multi-turn, context-aware interactions.

Key Features of Prompt Templates

Dynamic Content

Allows for variable input to tailor the prompt to specific tasks

Consistent Formatting

Ensures consistent prompt structure for predictable outputs

Ease of Experimentation

Facilitates quick testing of different prompt variations

Reusability

Versatile and suitable for diverse business applications

Customization

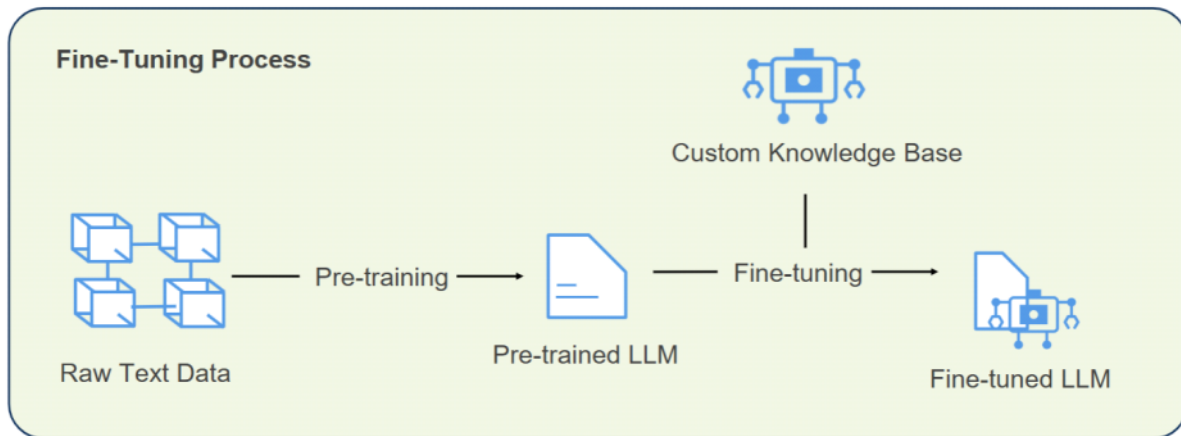
Enables tailoring prompts to specific applications and scenarios



Instruction Tuning Basics

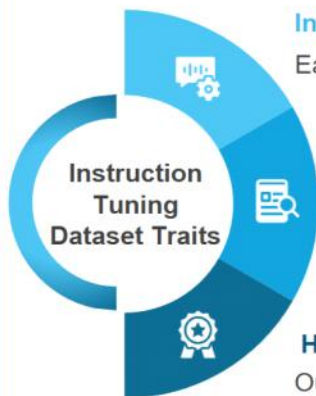
What is Fine-Tuning?

Fine-tuning involves training a pre-trained language model on a smaller, task-specific dataset to improve its performance on that particular task.



What is Instruction Tuning?

Instruction tuning is a fine-tuning process where a pre-trained LLM is further trained on a labeled dataset of instructional prompts and their corresponding outputs.



Instruction-output Pairs:

Each entry has a natural language instruction and its desired response.

Human-readable Input:

Instructions are in plain language for easy understanding.

High-quality Responses:

Outputs must be accurate and well-structured to ensure proper learning.



How Instruction Tuning Works

Step 1

Data Collection

Gather varied instruction response pairs to build a quality training set.



Step 2

Model Fine-Tuning

Train the model using supervised learning to map accurately to instructions.



Step 3

Evaluation and Iteration

Assess performance and refine the process to improve results.



Why Instruction Tuning Matters?



Task Alignment: Aligns prediction with instructions.



Efficient Learning: Improves zero/few-shot tasks.



Controlled Outputs: Ensures reliable responses.



Multi-Tasking: Supports various NLP tasks.

Benefits of Instruction Tuning



Reduces Prompt Complexity

Enables accurate results with simpler prompts.



Enhances Generalization

Effectively handles a broader range of instructions.



Improves Reliability

Delivers consistent outputs aligned with user intent.

Challenges and Limitations

Data Quality: Requires diverse, high-quality instruction response pairs for effective results.



Generalization: Models often struggle with unfamiliar instructions unlike those in training data.



Learning Depth: Tends to enhance response formatting more than deep understanding.



Compute Cost: Requires more resources than prompt engineering, even with optimizations.



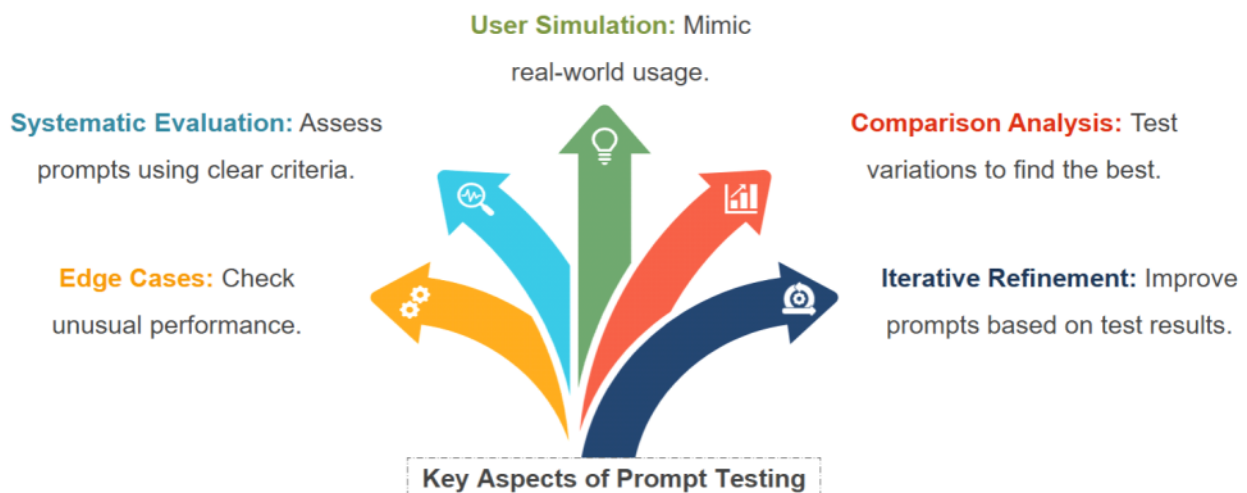
Instruction Tuning vs. Other Techniques

Aspect	Instruction Tuning	Fine-Tuning	Prompt Tuning
Data Requirement	Requires instruction-response paired datasets	Task-specific labeled data	Uses prompts without modifying model weights
Flexibility	Highly flexible and generalizes well across tasks	Limited flexibility, focused on trained tasks	Flexible but depends on prompt quality
Generalization	Performs well on unseen tasks	Performs best on tasks seen during training	Can attempt new tasks through prompt design
Resource Demand	Requires substantial computational resources	Moderate resource requirements	Minimal resources needed
Typical Use Cases	Virtual assistants and multi-task applications	Specialized domain models	Rapid prototyping and zero-shot learning

Prompt Testing and Evaluation

What is Prompt Testing?

Prompt testing is a systematic technique for evaluating and validating the effectiveness of prompts in AI interactions.



Methods of Prompt Testing



A/B Testing: Compare prompt variations for performance.

Stress Testing: Test prompts under extreme or complex conditions.

Semantic Analysis: Check response relevance and coherence.

User Feedback: Collect insights from real users.

Automated Testing: Run large-scale tests using tools.

Cross-Model Testing: Evaluate prompts across different models.

Scenario-Based Testing: Use targeted use cases to assess prompt quality.

Why Are Prompt Evaluations Important?



Improving Accuracy

Ensuring correct and relevant responses.



User Experience
Matching AI output with user expectations.



Reducing Hallucinations
Avoiding false or misleading information.



Workflow Optimization
Improving task efficiency and output quality.

Prompt Pitfalls and Debugging

Common Prompt Pitfalls

Ambiguity: Unclear prompts can cause irrelevant results.



Missing Context: Without background info, responses may be incomplete.

Too Specific: Overly narrow prompts may miss the bigger picture.



Token Limit: Long inputs may get cut off, losing key details.

Bias in Prompts: Poor wording or examples can skew the output.



Debugging Strategies

Clarify Prompts: Be specific to avoid confusion.

Test and Improve: Iterate, test, and document changes.

Check Format: Ensure responses are well-structured.

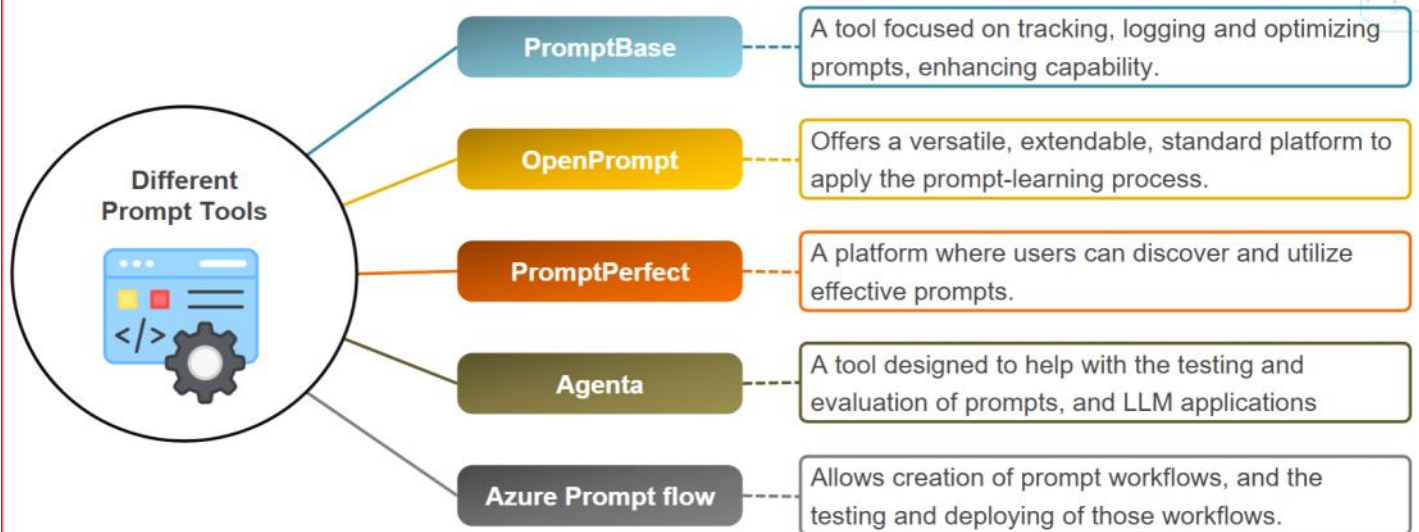


Use Good Examples: Add quality, relevant examples.

Watch for Bias: Review and refine to avoid biased outputs.

Collaborate: Share insights and keep records for future use.

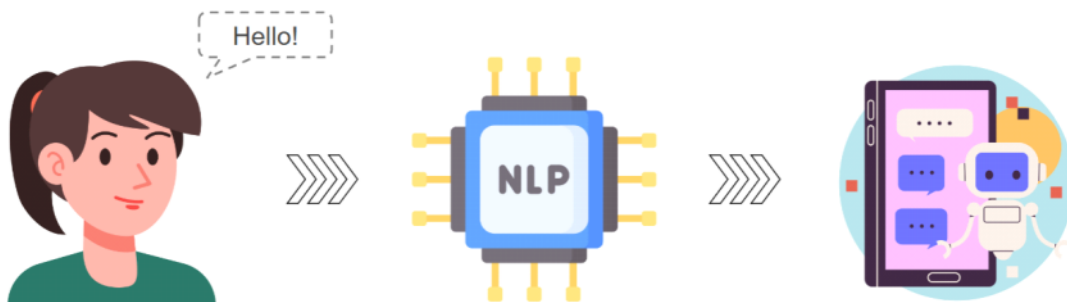
Different Prompt Tools



Prompts for Different NLP Tasks

What is Natural Language Processing?

Natural Language Processing (NLP) is a subfield of artificial intelligence that focuses on enabling computers to understand, interpret, and generate human language.



Text Summarization

Craft prompts that guide the model to summarize documents with a specified level of detail.



ChatGPT ▾

Saved memory full ⓘ

↑ Share

Natural Language Processing is a field of Artificial Intelligence that focuses on the interaction between computers and humans using natural language. It involves techniques such as tokenization, parsing, and sentiment analysis.

Summarize the paragraph in one sentence



Natural Language Processing is a branch of AI that focuses on enabling computers to understand and interpret human language using techniques like tokenization, parsing, and sentiment analysis.



Ask anything

+ Tools



Code Generation

Producing code snippets in response to a natural language instruction.

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↑ Share ... ⓘ

Write a one-line Python function that checks if a number is even.

Here's a one-line Python function to check if a number is even:

python

Copy

```
is_even = lambda x: x % 2 == 0
```

Example usage:

python

Copy

```
print(is_even(4)) # Output: True
print(is_even(5)) # Output: False
```



Question Answering



ChatGPT ▾

Share ...



The Hubble Space Telescope was launched in 1990 and remains in operation. Positioned above Earth's atmosphere, it avoids atmospheric distortion and captures high-resolution images. It has contributed to discoveries about the age of the universe, dark energy, and the life cycles of stars.

Question: When was the Hubble Space Telescope launched?

The Hubble Space Telescope was launched in **1990**.



Structuring prompts with specific instructions and context leads to a **concise** and **accurate** response.

Machine Translation

Well-crafted prompts guide the model to ensure **accurate**, **formal**, or **idiomatic** translations as needed.

ChatGPT ▾

Share ...



Translate the following English sentence into Korean :
"The weather is beautiful today."

The translation of "The weather is beautiful today." into Korean is:

"오늘 날씨가 아름다워요."



Text Classification



The act of predefined categories or labels to a given text based on its content, such as **sentiment analysis** or **topic identification**.

ChatGPT ▾

↑ Share ...



You're a sentiment detective. Analyze the mood of this message and label it as Positive, Negative, or Neutral: 'I've seen better service, but the food was decent.'

The mood of the message is **Neutral**. While the speaker mentions that the service could have been better, they also note that the food was decent, indicating a balanced or mixed perspective.



Data Extraction

ChatGPT ▾

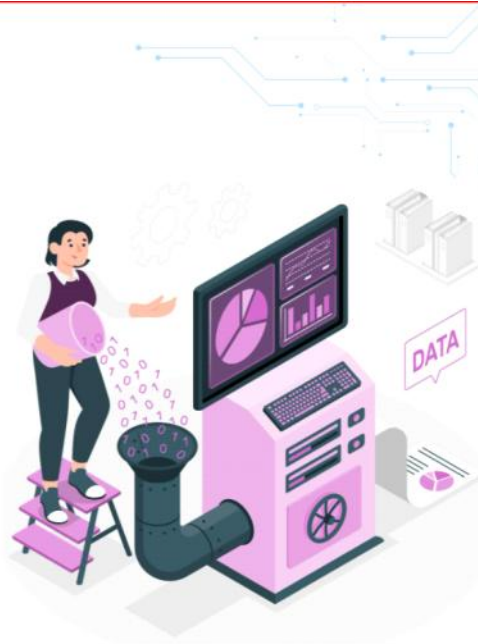
↑ Share ...



You are an AI assistant for a travel agency. Extract the travel destination, departure date, and preferred airline from this casual conversation:
'Hey! I'm thinking of flying to Tokyo sometime around July 10th. I'd love to fly with Emirates if possible. Can you check if there's anything available?'

Here are the extracted details from the conversation:

- **Travel Destination:** Tokyo
- **Departure Date:** Around July 10th
- **Preferred Airline:** Emirates



Demo 2: Designing Creative Prompts for Diverse NLP Tasks

Understanding Model Behavior with Prompt Variations

Prompt Variations Influence Output

ChatGPT

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List all programming languages mentioned in the text below.
Text: Python and JavaScript are commonly used in web development, while C++ is preferred for system programming.

The programming languages mentioned in the text are:

1. Python
2. JavaScript
3. C++

Ask anything

Tools

Different ways of **phrasing, structuring, or contextualizing prompts** can significantly change how AI models interpret tasks and generate responses.

Prompt Variations Influence Output

ChatGPT

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Share



Even subtle changes in wording or format can lead to variations in **accuracy, tone, and relevance.**

What technologies are mentioned in this text?
Text: "Python and JavaScript are commonly used in web development, while C++ is preferred for system programming."

The technologies mentioned in the text are:

1. Python (programming language)
2. JavaScript (programming language)
3. C++ (programming language)
4. Web development (technology domain)
5. System programming (technology domain)

Ask anything

Tools

Best Practices for Exploring Prompt Variations

Variations

Experiment with various prompt formats to improve output quality.

01

Clarity

Use clear, specific, and context-rich instructions.

02

Refinement

Refine prompts iteratively for optimal performance.

03

Evaluation

Evaluate responses for accuracy and Potential bias.

04

Demo 3: Refining Prompt Structure Through Iterative Debugging

Summary

In this lesson, you have learned to:

- e! Identify the structure and components of well-formed prompts for NLP tasks.
- e! Understand the application of zero-shot, one-shot, and few-shot prompting methods.
- e! Apply instruction tuning to improve model outputs and task relevance.
- e! Analyze prompt behavior using evaluation and identify areas for improvement.
- e! Evaluate prompt effectiveness and design pitfalls across various use cases.
- e! Create tailored and optimized prompts using strategic design approaches and tools.